

7.2 Hydrocarbons (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	7. Organic Chemistry (A Level Only)
Topic	7.2 Hydrocarbons (A Level Only)
Difficulty	Easy

Time allowed: 20

Score: /16

Percentage: /100

Question 1a

Benzene undergoes substitution reactions. State the equation for the reaction of benzene with nitric acid to produce nitrobenzene and water.

[2 marks]**Question 1b**

The structure of methylbenzene is shown below in Fig. 1.1.

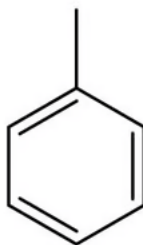


Fig. 1.1

Draw the structures of the two isomers of chloromethylbenzene formed from the reaction of methyl benzene and Cl_2 in the presence of AlCl_3 .

[2 marks]**Question 1c**

State the type of reaction that benzene will typically undergo.

[1 mark]

Question 2a

This question is about the bromination of benzene.

State the name of the mechanism that benzene will undergo if reacted with bromine in the presence of a halogen carrier to form bromobenzene, $\text{C}_6\text{H}_5\text{Br}$.

[1 mark]

Question 2b

Write an equation to show how the halogen carrier generates the Br^+ ion in order to allow the reaction in part (a) to occur.

[2 marks]

Question 2c

Complete the mechanism in Fig. 2.1 for the formation of bromobenzene. Include relevant charges where appropriate.

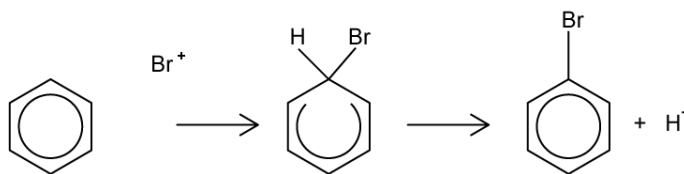


Fig. 2.1

[3 marks]

Question 3a

This question is about the reactions of methylbenzene.

Draw the structure of compound **A** in Fig. 3.1.

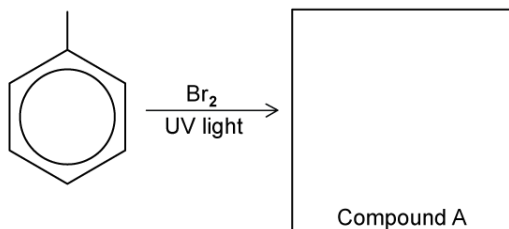


Fig. 3.1

[1 mark]

Question 3b

Draw the structure of compound **B** in Fig. 3.2.

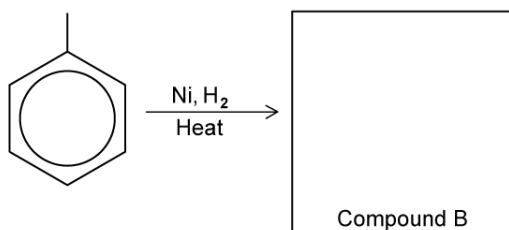


Fig. 3.2

[1 mark]

Question 3c

State the reagents and conditions required for reaction **1** in Fig. 3.3 for the formation of benzoic acid from methylbenzene.

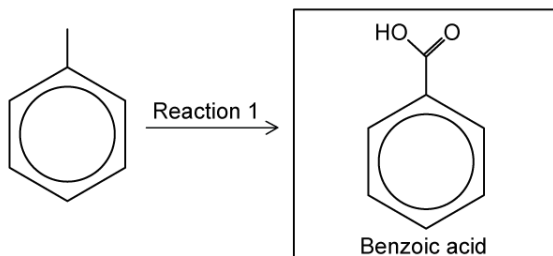


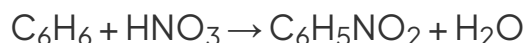
Fig. 3.3

[3 marks]

Model Answers: Easy

1a

a) The equation for the reaction of benzene with nitric acid to form nitrobenzene and water is:



- Correct formulas for reactants; [1 mark]
- Correct formulas for products; [1 mark]

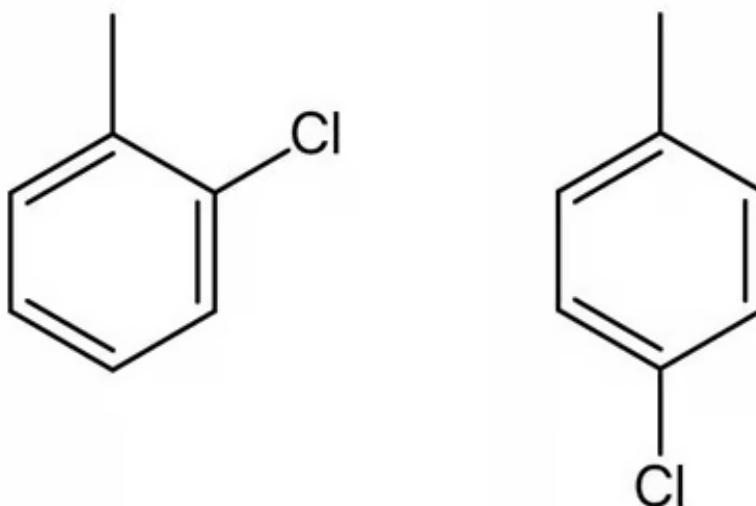
[Total: 2 marks]

- In this reaction, a nitro ($-\text{NO}_2$) group replaces a hydrogen atom on the arene
- The benzene is reacted with a mixture of concentrated nitric acid (HNO_3) and concentrated sulfuric acid (H_2SO_4) at a temperature between 25 and 60 °C
- Concentrated sulfuric acid acts as a catalyst and is regenerated at the end of the reaction

1b

b) The structures of the two isomers of chloromethylbenzene formed from the reaction of methyl benzene and Cl_2 in the presence of AlCl_3 are:

- Correct structure for 2-chloromethylbenzene; [1 mark]
- Correct structure for 4-chloromethylbenzene; [1 mark]



[Total: 2 marks]

- Alkylarenes such as methylbenzene undergo halogenation on the 2 or 4 positions
- This is due to the electron-donating alkyl groups which activate these positions
- If excess halogen is used then multiple substitution can occur in the 2, 4 and 6 positions
- **Remember:** The carbon atom in the benzene ring bonded to the methyl group is carbon number 1

1c

c) The type of reaction that benzene will typically undergo is:

- Electrophilic substitution; [1 mark]

[Total: 1 mark]

- There are three types of substitution reactions you should know: free radical, electrophilic, and nucleophilic
 - Free radical substitution occurs between alkanes and halogens
 - Electrophilic substitution occurs between benzene and nitric acid
 - Nucleophilic substitution occurs between halogenoalkanes and sodium

hydroxide

2a

a) The type of reaction that benzene will undergo is:

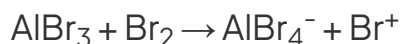
- Electrophilic substitution; [1 mark]

[Total: 1 mark]

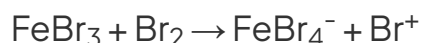
- The nature of benzene is different to other unsaturated compounds such as alkenes, so halogenation via electrophilic addition is not possible as it would involve breaking apart the stabilising ring of delocalised electrons
- Therefore benzene is most likely to undergo electrophilic substitution
- This mechanism restores the delocalisation in the benzene ring

2b

b) The equation to show how the halogen carrier generates the Br^+ ion is:



OR



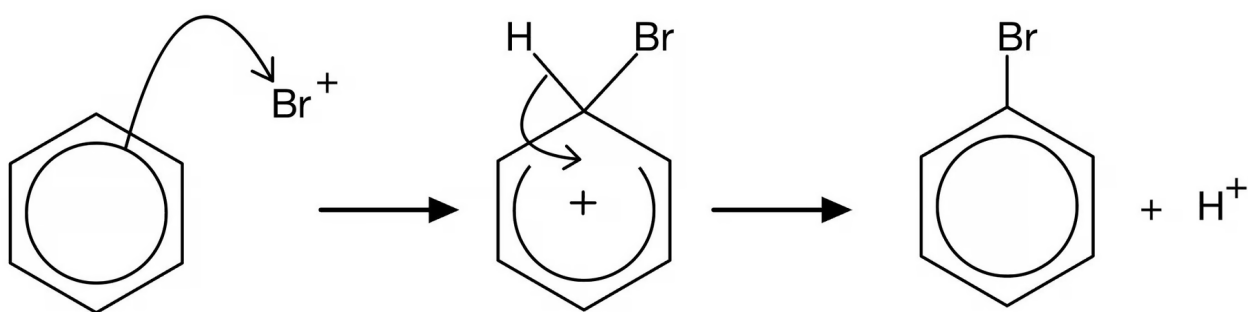
- Correct reactants; [1 mark]
- Correct products; [1 mark]

[Total: 2 marks]

- As the mechanism must be electrophilic substitution, the reaction must involve an electrophile
 - This is the Br^+ ion
- A halogen carrier needs to generate this by accepting one of the bromine atoms in the bromine molecule
- The halogen carrier is regenerated at the end of the reaction
 - $\text{AlBr}_3 + \text{Br}_2 \rightarrow \text{AlBr}_4^- + \text{Br}^+$
 - $\text{AlBr}_4^- + \text{H}^+ \rightarrow \text{AlBr}_3 + \text{HBr}$

2c

c) The completed mechanism should be:



- Curly arrow from benzene ring to Br^+ ; [1 mark]
- Positive charge in intermediate; [1 mark]
- Curly arrow from C-H bond to regenerate benzene ring in intermediate; [1 mark]

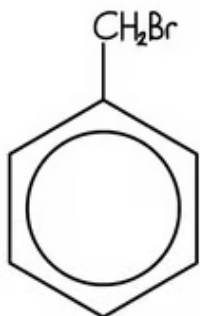
[Total: 3 marks]

- The question asks you to draw any relevant charges - this is the positive charge in the intermediate
- The electrophile is bonded to the benzene ring, but so still is the H atom
- The delocalised system is temporarily broken to leave a partially delocalised electron system and a positive charge
- This is then restored by the movement of electrons from the C-H bond to the benzene ring

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3a

a) The structure of compound **A** is:



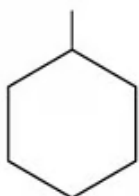
- Correct structure of compound **A**; [1 mark]

[Total: 1 mark]

- The methyl group is a side chain of the benzene ring
- Halogenation of alkylarenes in the side chain will occur when the halogen is passed into boiling alkylarene in the presence of ultraviolet (UV) light
- This is a free-radical substitution reaction
- If a halogen carrier (AlBr_3 or FeBr_3) was present then a substitution reaction would occur on the benzene ring (electrophilic substitution)

3b

b) The structure of compound **B** is:



- Correct structure of compound **B**; [1 mark]

[Total: 1 mark]

- Methylbenzene is reduced to methylcyclohexane in this reaction
- This is also known as hydrogenation

3c

c) The reagents and conditions required are:

- Alkaline potassium manganate(VII) / KMnO_4 ; [1 mark]
- Heat under reflux; [1 mark]
- Acidified by dilute sulfuric acid; [1 mark]

[Total: 3 marks]

- Alkyl groups are usually fairly resistant to oxidation unless they are attached to a benzene ring
- They then are easily oxidised by an alkaline solution of potassium manganate(VII)
- Oxidation usually requires acidified potassium manganate(VII) but in this case alkaline conditions using sodium carbonate are required followed by acidification with dilute sulfuric acid
- In this reaction the purple colour of the potassium manganate(VII) is eventually replaced by a dark brown precipitate of manganese(IV) oxide

